WORKED SOLUTION

STEP I 2000, Question 7

Pure • Bounding a Derivative

> **STEP I 2000, Question 7**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/00-s1-q7

The key idea

Differentiate, and the condition $\mathrm{f}'(x) \geq 0$ becomes "a is at least as big as a certain function of x". So everything hinges on the *largest* value of that function. It depends only on x^2, and after the substitution $u = 1 + x^2$ it becomes a quadratic in $1/u$. Completing the square then gives the maximum, and shows exactly where it occurs.

Solution

The derivative

By the quotient rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^3}{1+x^2}\right) = \frac{3x^2(1+x^2) - x^3 \cdot 2x}{(1+x^2)^2} = \frac{x^4 + 3x^2}{(1+x^2)^2},$$

so

$$\mathrm{f}'(x) = a - g(x), \qquad \text{where } g(x) = \frac{x^2(x^2+3)}{(1+x^2)^2}.$$

Hence $\mathrm{f}'(x) \geq 0$ for all x exactly when $a \geq g(x)$ for all x. We need the largest value of g.

The largest value of g

Put $u = 1 + x^2$, so that $u \geq 1$ and $x^2 = u - 1$. Then

$$g = \frac{(u-1)(u+2)}{u^2} = \frac{u^2 + u - 2}{u^2} = 1 + \frac{1}{u} - \frac{2}{u^2}.$$

? **Ask yourself** Why is it a good idea to write this in terms of $1/u$ rather than u?

In terms of $s = 1/u$, $g = 1 + s - 2s^2$, which is just a quadratic, and the maximum of a

quadratic comes from completing the square, with no calculus needed:

$$1 + s - 2s^2 = \frac{9}{8} - 2\left(s - \frac{1}{4}\right)^2.$$

Since a square is never negative, $g \leq \frac{9}{8}$ for every x, with equality only when $s = \frac{1}{4}$, that is $u = 4$, $x = \pm\sqrt{3}$.

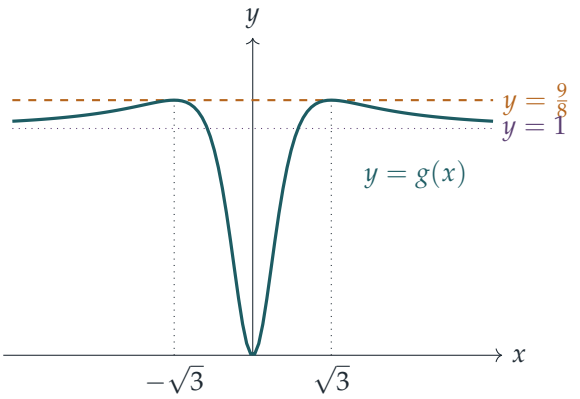
Putting this back in terms of x (using $s - \frac{1}{4} = \frac{3-x^2}{4(1+x^2)}$) gives a tidy identity for the derivative:

$$\mathrm{f}'(x) = \left(a - \frac{9}{8}\right) + \frac{(3-x^2)^2}{8(1+x^2)^2}.$$

Conclusion

If $a \geq \frac{9}{8}$, the first bracket is non-negative and the second term is a square divided by something positive, so it is non-negative too. Therefore $\mathrm{f}'(x) \geq 0$ for all x, as required.

The constant $\frac{9}{8}$ is also the best possible: if $a < \frac{9}{8}$ then $\mathrm{f}'(\sqrt{3}) = a - \frac{9}{8} < 0$. When $a = \frac{9}{8}$, f' is zero only at $x = \pm\sqrt{3}$, so f is still strictly increasing, with two stationary points of inflection.



The graph of g climbs from 0, overshoots its limit 1, peaks at $\frac{9}{8}$ when $x = \pm\sqrt{3}$, and then falls back towards 1. The derivative $\mathrm{f}'(x)$ is the height of the line $y = a$ above this curve.

Checks

- At $x = \sqrt{3}$: $g = \dfrac{3 \cdot 6}{4^2} = \dfrac{18}{16} = \dfrac{9}{8}$. ✓
- As $x \to \infty$, $g \to 1 < \frac{9}{8}$. This makes sense, since $\frac{x^3}{1+x^2}$ behaves like x for large x, so its slope tends to 1.

- Numerically, with $a = \frac{9}{8}$ the smallest value of $\mathrm{f}'(x)$ on a grid of step 0.001 over $[-100, 100]$ is about 2×10^{-10} (essentially 0, at $x \approx \pm\sqrt{3}$). With $a = 1.1$, $\mathrm{f}'(\sqrt{3}) = -0.025 < 0$.

Common slips

- Treating g's limit at infinity, 1, as its largest value. The curve overshoots, which is why the answer is $\frac{9}{8}$ and not 1.
- Differentiating g directly and getting lost in the algebra. The substitution $s = 1/(1+x^2)$ turns it into a one-line completion of the square.

Beyond STEP

Here a "for all x" statement was reduced to finding a *supremum*: $\mathrm{f}' \geq 0$ everywhere exactly when $a \geq \sup g$. The fact that g has a limit at infinity but is largest at a finite point is a good warning that a limit need not be a bound. In analysis you'll prove that a function with non-negative derivative is increasing (via the Mean Value Theorem), which is why this question is really about whether f is monotonic. Abbott's *Understanding Analysis* develops these ideas carefully.

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